



QuantEM: The Quantum Error Management Compiler

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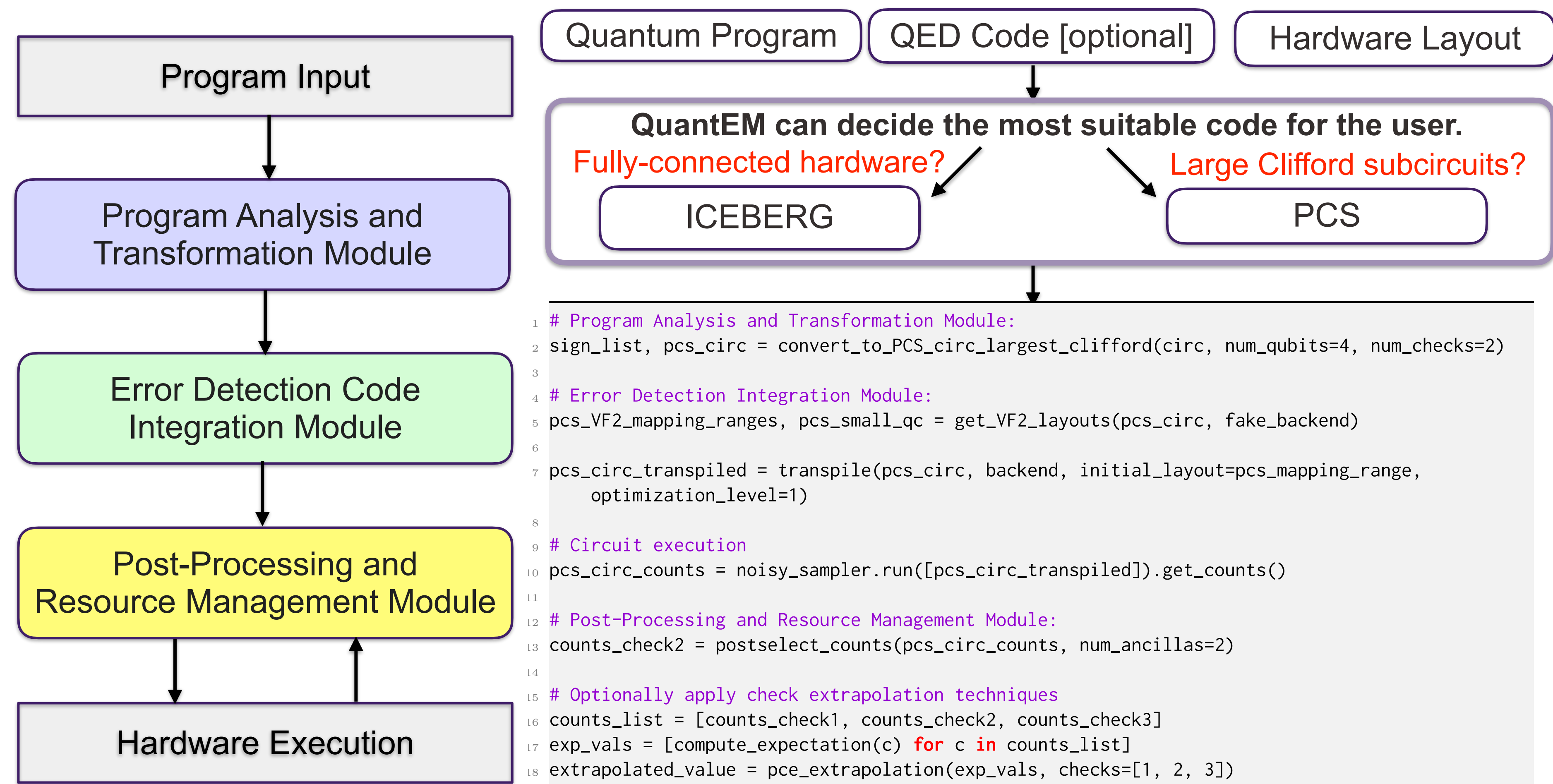
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Compilation for QED Codes

As quantum computers advance towards fault-tolerant architectures, Quantum Error Detection (QED) codes emerge as an intermediate strategy for improving the reliability and utility of quantum programs.

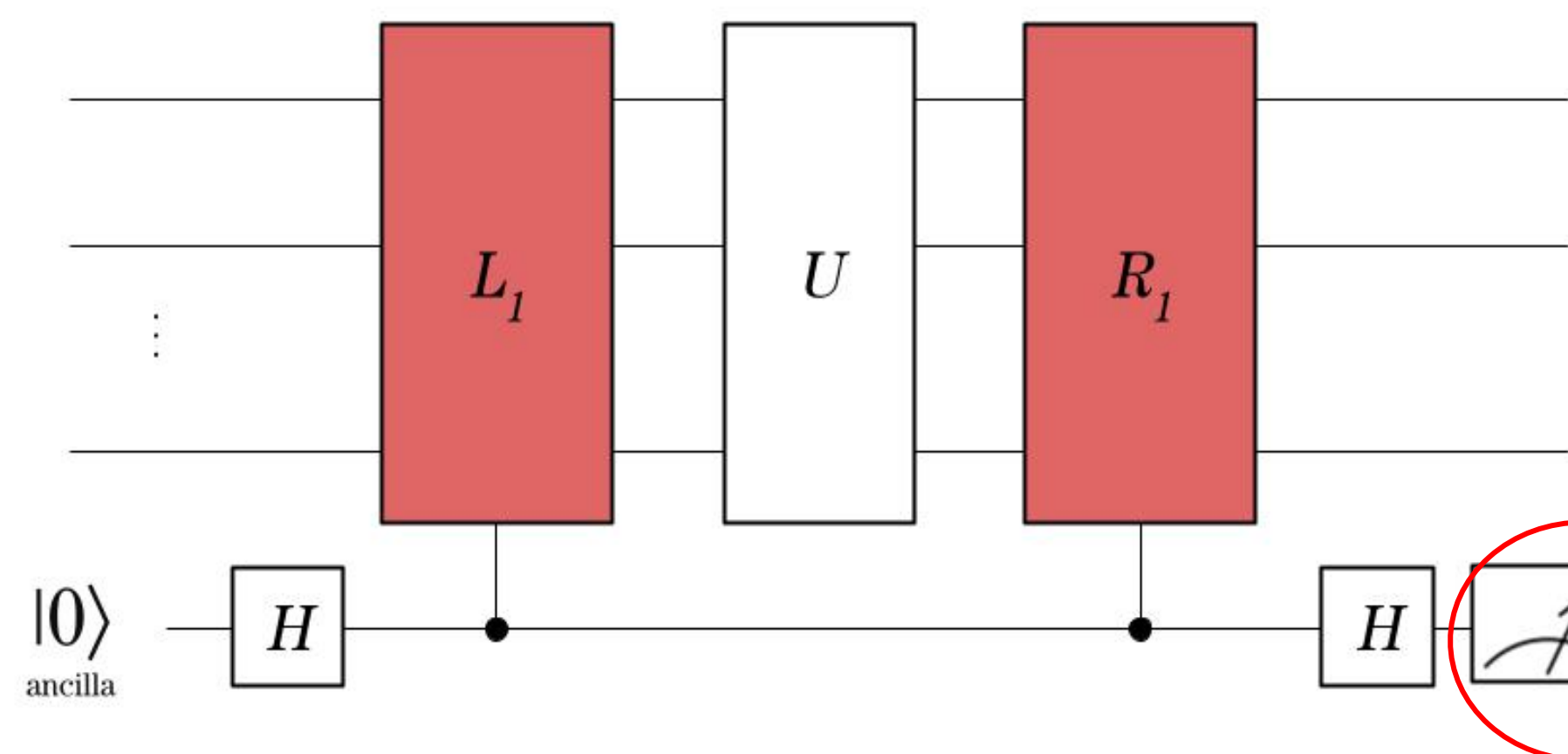
- Applying QED to arbitrary quantum circuits remains challenging because of the need for manual insertion of detection subcircuits, ancilla allocation, and hardware-specific mapping and scheduling.
- Solution:** We present **QuantEM**, a modular and extensible compiler that automates the insertion of QED codes to arbitrary quantum programs.

QuantEM Workflow



Background: Pauli Check Sandwiching

- QuantEM currently supports **Pauli Check Sandwiching (PCS)**¹, **ancilla-free PCS**², and **ICEBERG**³ codes.
- PCS** surrounds a payload unitary circuit with left and right controlled Pauli 'checks'.



Pauli checks anti-commute with errors that occur in the payload circuit, causing detectable phase kickback in ancillas.

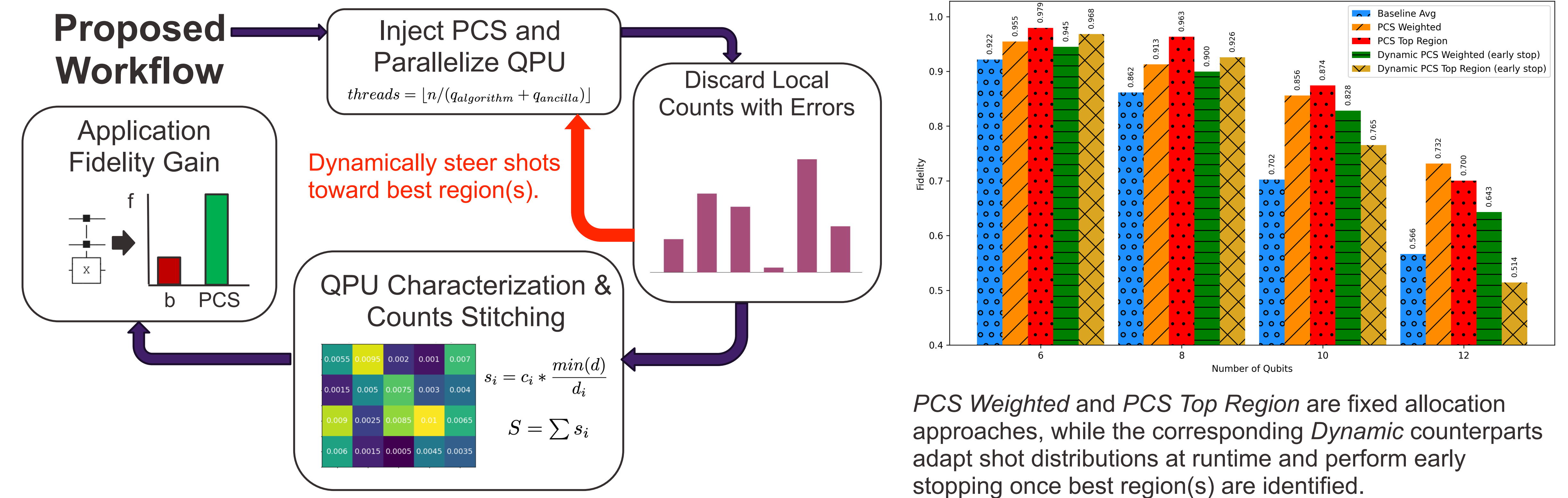
Measure 1 → Discard
Measure 0 → Keep

References

- [1] Gonzales, Alvin, et al. "Quantum Error Mitigation by Pauli Check Sandwiching." Scientific Reports 13.1 (2023): 2122.
- [2] Quinn Langfitt, Alvin Gonzales, Joshua Gao, Ji Liu, Zain H. Saleem, Nikos Hardavellas, and Kaitlin N. Smith. Dynamic resource allocation with quantum error detection, 2024.
- [3] Chris N Self, Marcello Benedetti, and David Amaro. Protecting expressive circuits with a quantum error detection code. Nature Physics, 20(2):219–224, 2024
- [4] Langfitt, Quinn, et al. "Extrapolating Pauli Checks for Expectation Value Estimation on Noisy Quantum Devices." arXiv preprint arXiv:2406.14759 (2025).

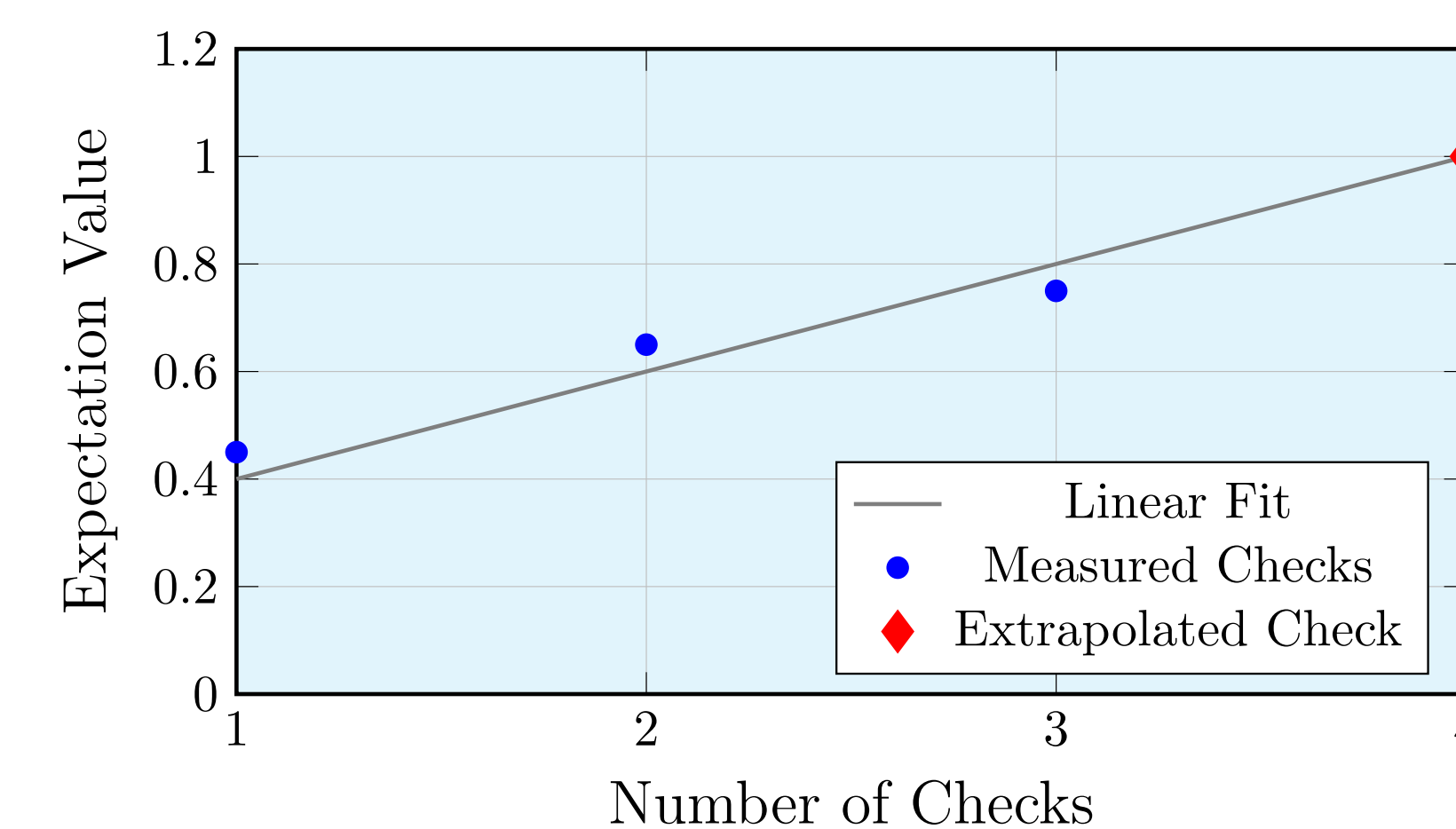
Use Case: Dynamic Error Characterization and Qubit Mapping

Quantum processors can have varying levels of noise across the chip. Thus, error characterization is crucial for effective utilization of the chip. Additionally, we incorporate the Upper-Confidence Bound (UCB) algorithm that intelligently adapts resource distribution based on real-time acceptance rates from Pauli checks, enabling significant shot budget savings (up to 58% savings for QAOA).²

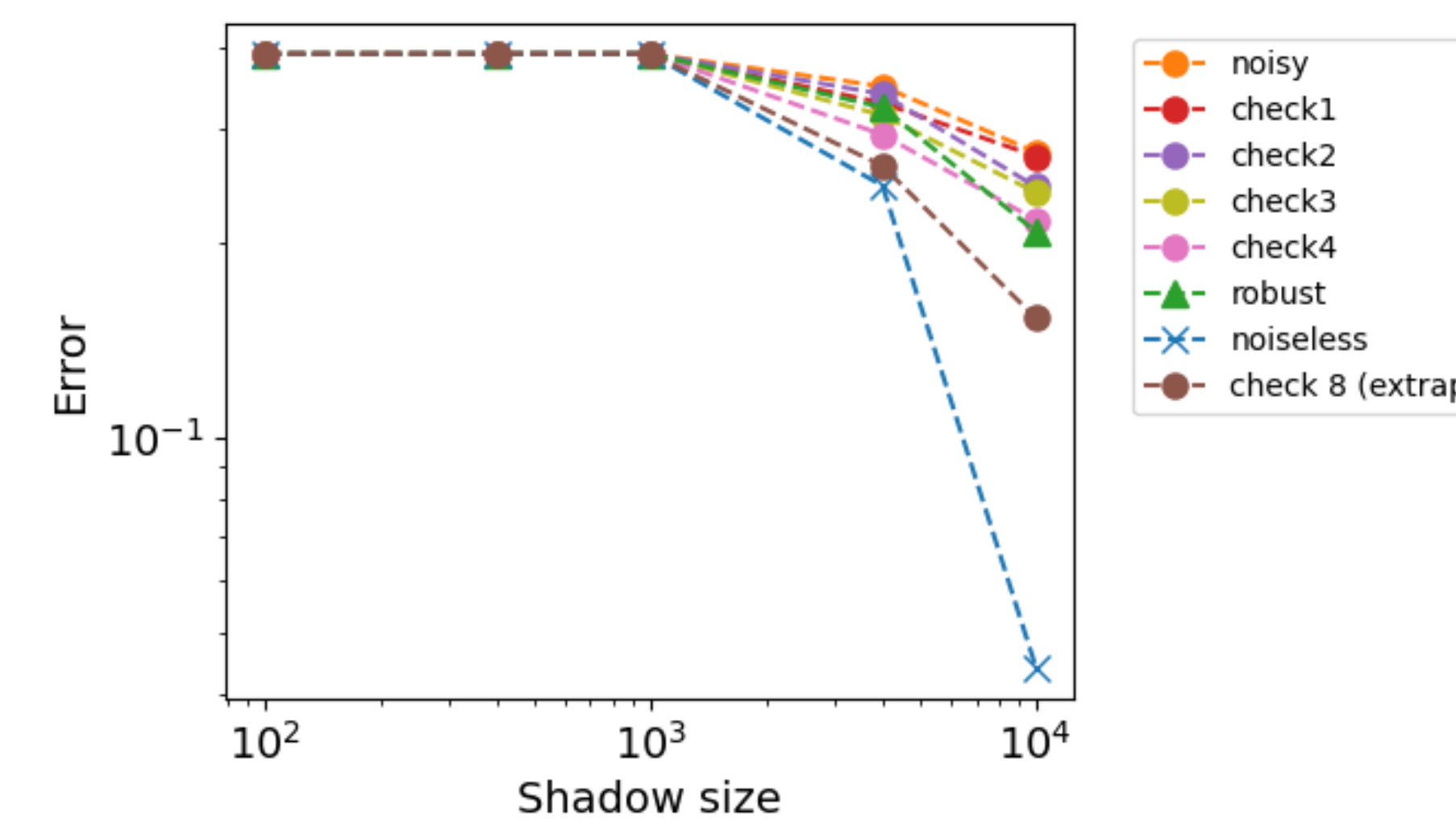


Use Case: Post-Processing Module for Error Detection Extrapolation

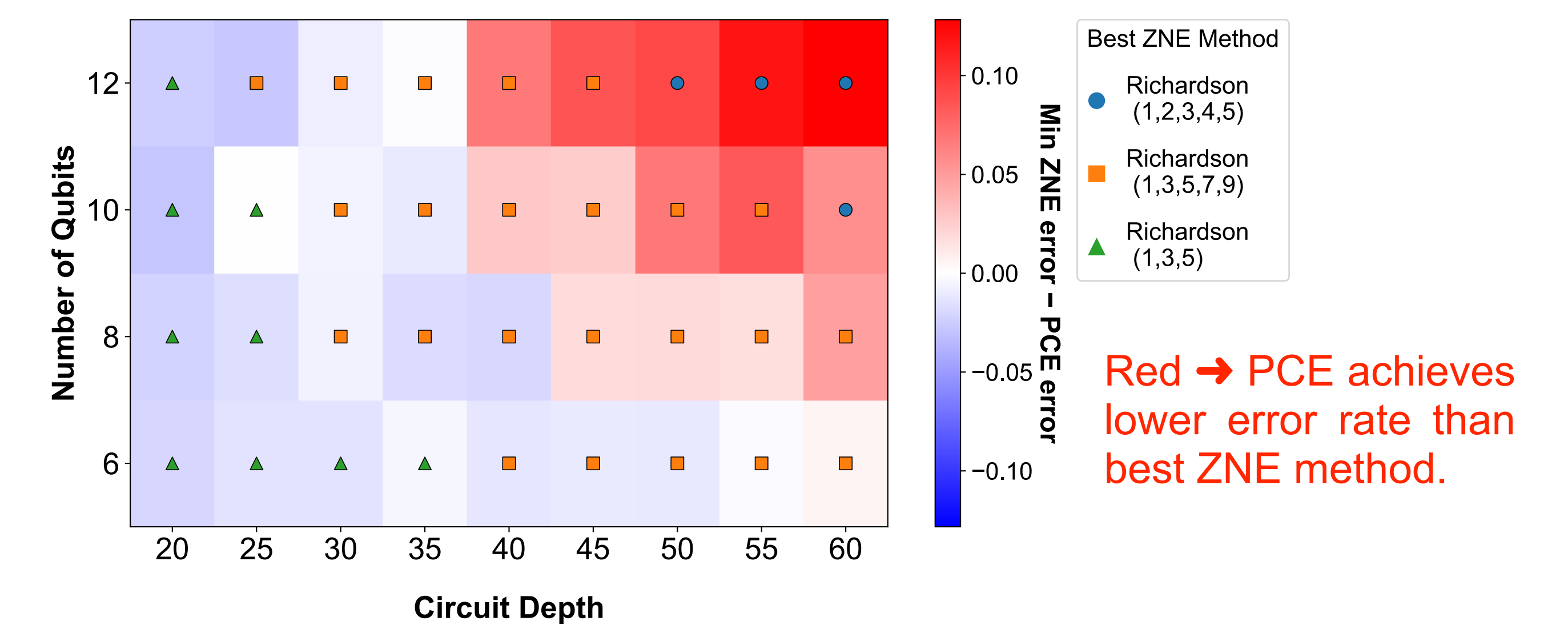
Based on the number of ancilla qubits and observed detection patterns, can optionally apply check extrapolation techniques to estimate the final expected outcome. Below we show results for various Pauli Check Extrapolation (PCE)⁴ experiments. PCE is an error mitigation scheme that obtains measured expectation values from circuits with different numbers of checks and extrapolates to the "maximum check" limit.



QuantEM can extrapolate measured expected values to the theoretical ideal limit.



Error mitigating classical shadows: PCE vs. Robust Shadow estimation when extrapolating from 4 to 8.



Benchmark comparison of PCE against ZNE on random Clifford circuits with depolarizing noise. Results were averaged over 20 random circuits.

Future Updates

- Extend Pauli Check Sandwiching support to qudit-based quantum systems.
- Expand compatibility to additional hardware vendors beyond Qiskit and superconducting platforms.
- Include use cases and optimizations for Iceberg codes, such as error characterization for trapped-ion systems.
- Integrate new error detection codes as they become available.